

s_1	s_2	s_3	s_4	s_5
+5	+2	0	+1	+10

$$a = \{L, R\}$$

s_1 and s_5 are terminal

$$Q(s, a) \leftarrow Q(s, a) + \alpha (R_{t+1} + \gamma \max_a (Q(s', a')) - Q(s, a))$$

$$Q(s_1, L)$$

$$Q(s_1, R)$$

$$Q(s_2, L)$$

$$Q(s_2, R)$$

$$\pi(a|s)$$

$$\pi(R|s_3)$$

small chance
of random
action

greedy policy or ϵ -greedy

$$Q(s_3, L) \quad Q(s_3, R)$$

o	o	o	o	o	o	o
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	s_1	s_2	s_3	s_4	s_5	
$t_0:$	0	0	0	0	0	$Q=0.5$ $\gamma=0.8$

The agent moves L twice

$$\begin{aligned}
 Q(s_3, L) &\leftarrow Q(s_3, L) + \alpha (R_{t+1} + \gamma \max_a (Q(s_2, a_{t+1})) - Q(s_3, L)) \\
 &\leftarrow 0 + 0.5 (+2 + 0 - 0) \\
 &\leftarrow 1
 \end{aligned}$$

0	0	0	0	1	0	0	0	0	0
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always 0
for terminal
states

$$\begin{aligned}
 Q(s_2, L) &\leftarrow 0 + 0.5 [+5 + 0 - 0] \\
 &\leftarrow 2.5
 \end{aligned}$$

0	0	2.5	0	1	0	0	0	0	0
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Trial 2: Agent goes R twice

from $\pi(a|s)$ is ϵ -greedy

$$\max_a (Q(s_4, a_{t+1})) = 0$$

$$Q(s_3, R) \leftarrow 0 + 0.5(+1 + 0 - 0) \leq 0.5$$

0	0	2.5	0	1	0.5	0	0	0	0
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$$Q(s_4, R) \leftarrow 0 + 0.5(10 + 0 - 0) \leq 5$$

$$\gamma \max_a (Q(s', a'))$$

0	0	2.5	0	1	0.5	0	5	0	0
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Trial 3: Agent chooses L, R, R, R

$$Q(s_3, L) \leftarrow 1 + 0.5(+2 + 0.8(2.5) - 1) \leq 2.5$$

0	0	2.5	0	2.5	0.5	0	5	0	0
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$$Q(S_2, R) \leq 0 + 0.5(0 + 0.8(2.5) - 0) \\ \leq 1$$

0	0	2.5	1	2.5	0.5	0	5	0	0
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$$Q(S_3, R) \leq 0.5 + 0.5(1 + 0.8(5) - 0.5) \\ \leq 2.75$$

0	0	2.5	1	2.5	2.75	0	5	0	0
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$$Q(S_4, R) \leq 5 + 0.5(10 + 0 - 5) \\ \leq 7.5$$

0	0	2.5	1	2.5	2.75	0	7.5	0	0
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